Giggle Reloaded (GGRD) Whitepaper v4.0

Contract-aligned mainnet specification for tokenomics, governance, presale, staking, vesting, and operational controls

Version: v4.0 (contract-aligned rewrite)

Date: 2026-03-10

Chain: BNB Smart Chain Mainnet (chainId 56)

Primary source: `ggrd-bsc-contracts`

Deployment snapshot: `deployments/bscMainnet.latest.json`

Token: Giggle Reloaded (GGRD), fixed supply model

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1. Scope and Methodology

This whitepaper is a full rewrite based on deployed contract logic and repository specifications. It does not reuse prior narrative whitepaper structure. Every numeric parameter in this document is either contract-enforced or explicitly marked as an operational assumption from the repository docs.

Source hierarchy used in this document

- Level 1 `contracts/*.sol` for immutable and runtime-enforced rules.
- Level 2 `deployments/bscMainnet.latest.json` for live mainnet addresses and constructor timestamps.
- Level 3 `docs/SPEC_TOKENOMICS_v1.md` and architecture/security docs for declared launch assumptions.

When contract code and descriptive text differ, contract code is treated as canonical.

2. Project Vision and Design Constraints

GGRD is designed as a fixed-supply BSC token with deterministic on-chain sale mechanics, governance delay controls, staking emissions caps, and charity vesting logic. The design target is auditable execution with reduced discretionary control surface.

Explicit constraints

- Fixed supply token model.
- 0% transfer tax.
- No post-deployment mint API.
- No blacklist, no freeze, no confiscation logic.
- Governance via Safe multisig plus timelock layers.
- Presale escrow with success/fail settlement paths.

3. Token Contract and Monetary Policy

Field	Value	Contract anchor
Name	Giggle Reloaded	`GGRDToken.sol constructor`
Symbol	GGRD	`GGRDToken.sol constructor`
Decimals	18	`OpenZeppelin ERC20 default`
Total supply	10,000,000 GGRD minted once in constructor	`_mint(recipient, 10_000_000 ether)`
Transfer tax	No tax mechanism in contract	`GGRDToken.sol`
Post-deploy mint	No public or owner mint function	`GGRDToken.sol`

Field	Value	Contract anchor
Blacklist/freeze	No account restriction functions	`GGRDToken.sol`

Mint destination in mainnet deployment artifact is deployer address at deployment time; downstream allocations are performed by transfers to module contracts.

4. Allocation and Economic Model

The total supply model combines hard caps from contracts and launch allocations from repository specification.

Component	Amount (GGRD)	Share	Enforcement type
Presale token cap	3,000,000	30%	On-chain cap in `PresaleEscrow8020.tokenCap`
Staking reward funding cap	3,000,000	30%	On-chain cap in `GGRDStakingMultiV1.REWARD_POOL_CAP`
Charity target allocation	2,000,000	20%	Operational allocation from `docs/SPEC_TOKENOMICS_v1.md`
Operations and liquidity	2,000,000	20%	Operational allocation from `docs/SPEC_TOKENOMICS_v1.md`

CharityVesting403030 enforces distribution shape (40/30/30) and vesting schedule, but not a fixed token amount. The amount is determined by funding transfers.

5. Presale Escrow Mechanics

Presale is implemented in `PresaleEscrow8020.sol` as USDC escrow with deterministic stage logic, wallet caps, success split, refund fallback, and post-success claim vesting.

5.1 Core parameters (contract-enforced)

Parameter	Value	Anchor
Payment asset	USDC on BSC	`payToken` immutable
Duration	21 days	`PRESALE_DURATION`
Softcap	20,000 USDC	`softcap = 20_000 * PAY_UNIT`
Hardcap	31,250 USDC	`hardcap = 31_250 * PAY_UNIT`

Parameter	Value	Anchor
Min buy	50 USDC	`minBuy = 50 * PAY_UNIT`
Global wallet cap	3,000 USDC	`globalMaxPerWallet`
Total token cap	3,000,000 GGRD	`tokenCap = 3_000_000 * SALE_UNIT`
Claim vesting duration	60 days	`CLAIM_LINEAR_DURATION`

5.2 Stage matrix

Stage	Elapsed rule	Allocation (GGRD)	Price (USDC)	Wallet cap (USDC)	Max raise (USDC)
1	`elapsed < 5 days`	1,000,000	0.0085	500	8,500
2	`5-10 days`	800,000	0.0100	750	8,000
3	`10-16 days`	700,000	0.0115	1,000	8,050
4	`16-21 days`	500,000	0.0134	1,500	6,700

Stage max raise sum: 31,250 USDC, matching hardcap.

5.3 Mainnet schedule from deployment constructor

Milestone	UTC datetime	Source
Presale start	2026-03-10 01:15:19	`startTs = 1773105319` in deployment args
Stage 1 end	2026-03-15 01:15:19	start + 5 days
Stage 2 end	2026-03-20 01:15:19	start + 10 days
Stage 3 end	2026-03-26 01:15:19	start + 16 days
Presale end	2026-03-31 01:15:19	start + 21 days

5.4 Settlement and claim logic

- **Finalize success condition:** `totalRaised >= softcap`.
- **Success split:** 80% USDC to liquidity wallet and 20% USDC to marketing wallet.
- **Failure path:** each buyer can call `refund()` for 100% contributed USDC.
- **Success claim path:** buyers claim GGRD by vesting schedule, 50% immediate and 50% linear over 60 days.
- **Finalize timing gate:** owner can finalize only after end time or at hardcap.

Quote formula in contract: `tokenOut = payIn * PRICE_SCALE * SALE_UNIT / (stagePriceScaled * PAY_UNIT)`.

6. Governance and Control Plane

Governance model combines Safe multisig operations with timelock-enforced delays for selected high-impact actions.

Control layer	Current mainnet value	Implementation
Operational Safe	2 of 3 signers	External Safe setup referenced in deployment docs
Short timelock	48 hours (172800 seconds)	`GGRDTimelock` constructor args
Long timelock	72 hours (259200 seconds)	`GGRDTimelock` constructor args
Staking owner	Long timelock address	`GGRDStakingMultiV1` constructor owner arg
Staking guardian	Safe address	`GGRDStakingMultiV1` constructor guardian arg
Airdrop owner	Short timelock address	`MerkleAirdrop` constructor owner arg

7. Staking Engine

`GGRDStakingMultiV1` is a multi-position staking contract with reward and principal caps, lock-based reward eligibility, and early withdraw penalties.

Parameter	Value	Effect
APR	10% (`APR_BPS=1000`)	Linear annualized reward accrual
Minimum lock	30 days	No rewards before lock maturity
Program duration	730 days	Reward accrual stop at `endTime`
Reward funding cap	3,000,000 GGRD	Hard upper bound on funded rewards
Max total staked principal	5,000,000 GGRD	Hard principal capacity cap
Early withdraw penalty	10%	Penalty before 30-day lock, no reward payout

Parameter	Value	Effect
Emergency controls	guardian or owner can pause; owner unpauses	Operational incident response control

8. Charity Vesting Program

Charity release is executed by `CharityVesting403030` with deterministic share splitting and linear vesting after cliff.

Parameter	Value	Contract behavior
Split	40% / 30% / 30%	Encoded by `A_BPS/B_BPS/C_BPS`
Cliff behavior	No vesting before cliff	`if (timestamp < cliff) return 0`
Post-cliff behavior	Linear release until duration end	`(total * (timestamp - start)) / duration`
Mainnet start	2026-03-10 00:15:17 UTC	constructor arg `1773101717`
Mainnet cliff end	2027-03-10 00:15:17 UTC	`31536000` seconds
Mainnet full vesting end	2029-03-09 00:15:17 UTC	`94608000` seconds

Current deployment maps charity A/B/C recipients to one operational charity safe address.

9. Airdrop Distribution Framework

`MerkleAirdrop` supports multiple rounds with independent Merkle roots and claim windows.

- Round creation requires merkle root, start/end times, and funded amount.
- Claim authorization uses standard Merkle proof verification over `(index, account, amount)`.
- One-claim-per-leaf enforced through bitmap state.
- Owner can sweep unclaimed leftovers only after round expiry.

10. Liquidity and LP Lock Policy

Economic intent from repo specification is GGRD/USDC liquidity provisioning after successful presale finalization.

- Presale USDC success split sends 80% to liquidity operations wallet and 20% to marketing wallet.

- Specification reference launch ratio: 25,000 USDC paired with approximately 1,200,000 GGRD.
- LP lock target: 12 months.

LP lock is implemented by `LPLocker.sol` contract pattern (owner can extend lock only forward, and withdraw only after unlock time).

> LP token address does not exist until pair creation. Therefore LP locker deployment/lock execution is an explicit post-presale operational step.

11. Security Controls and Residual Risks

Implemented controls

- OpenZeppelin primitives across token, timelock, access control, Merkle verification, and safe transfers.
- Reentrancy protections in presale, staking, and airdrop state-changing flows.
- Hard caps for presale token sale, staking rewards, and staking principal.
- Pause control in staking with guardian-owner split to improve incident response speed.

Residual risks

- Misconfigured wallet addresses in deployment/env can route funds incorrectly.
- High-impact governance transactions still require strict signer discipline.
- Operational execution (LP provisioning/locking, funding schedules) remains process-sensitive.

12. Mainnet Addresses and Parameters

Addresses below are sourced from `deployments/bscMainnet.latest.json`.

Component	Address	Role
GGRD token	`0xA0d5663d57b7D7EF975D2F02BcAEaf5c94c671f9`	ERC20 fixed supply token
PresaleEscrow8020	`0xd8983534dd3c369d85127f6C9B85d98768139387`	USDC escrow presale
USDC pay token	`0x8AC76a51cc950d9822D68b83fE1Ad97B32Cd580d`	Presale payment asset
GGRDStakingMultiV1	`0x05d803135bc09e9E5006CA75C533377D7Ea0A34b`	Staking module
CharityVesting403030	`0xE2993F0C8D2020dD9866E2d6dC5AA1D62b7C8fDC`	Charity vesting splitter

Component	Address	Role
MerkleAirdrop	`0x9F64E78185ee697783E0c9ebA0F66131FFCedBD6`	Airdrop module
Timelock short (48h)	`0xBC4655E8D5A4A4B16bD30db2fFBC94Ed05AbcE87`	Short governance delay
Timelock long (72h)	`0xd4b5f94f7cF8Eb79bEd2f3Aa5693d9F97D573E31`	Long governance delay
Governance Safe (2/3)	`0x28218d08f6D2084eBd7CeF3f582139E7417b6300`	Operational multisig
Liquidity Safe (2/3)	`0x7677AC2712d161e7F75fC4316E8B3dC3FaC01009`	USDC destination on presale success (80%)
Marketing Safe (2/3)	`0xF2922343AdF6F2631A2f1e9b9b11c9d269f3874D`	USDC destination on presale success (20%)
Charity Safe A	`0xd58B8D79deC67F8731267B929D24A4923Bac59a3`	Current charity vesting recipient mapping

13. Operational Runbook and Governance Actions

Presale phase

- Fund presale contract with up to 3,000,000 GGRD.
- Monitor stage progress, wallet caps, and raised total against hardcap/softcap thresholds.
- Trigger `finalize()` after end time or at hardcap.

Post-finalize (success path)

- Execute LP provisioning workflow using liquidity wallet USDC share and planned GGRD allocation.
- Create LP pair token and deploy/initialize LP locker for 12-month lock target.
- Publish LP pair and locker addresses with explorer links.

Post-finalize (failure path)

- Keep refund interface open and monitored until participant refunds are completed.
- Communicate final accounting and remaining balances transparently.

Staking and charity lifecycle

- Fund staking reward pool gradually within `REWARD_POOL_CAP`.
- Initialize staking program start time under governance control.
- Call charity vesting release function per schedule and publish transfer reports.

14. Risk and Legal Notice

- GGRD is a high-risk crypto asset; total capital loss is possible.
- This document is technical information, not investment, legal, or tax advice.
- Users and operators should verify contract addresses and transaction payloads independently.
- Operational execution quality materially affects real-world outcomes despite strong contract controls.

Whitepaper v4.0 is a repository-grounded technical specification as of 2026-03-10. Future governance actions, module upgrades, or operational reconfiguration can change non-immutable behavior.